



Diffusion Model: Application, Method, and Implementation

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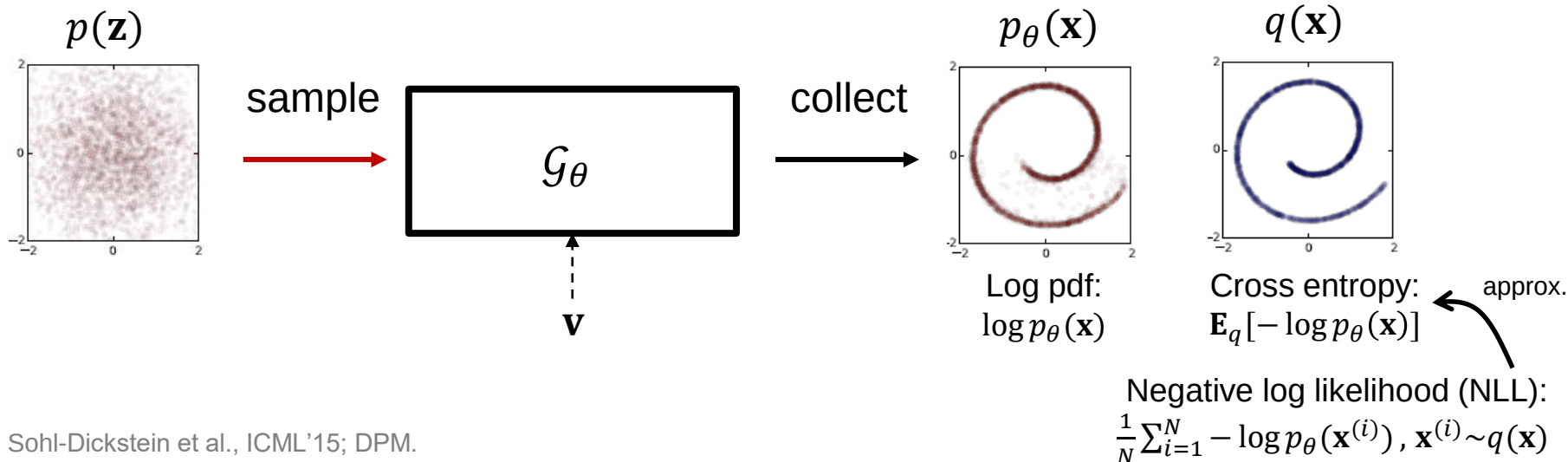


Outline

- **Application**
 - **Deep generative models**
 - Image synthesis
 - Image restoration and enhancement
 - Appendices for equations

Generative model

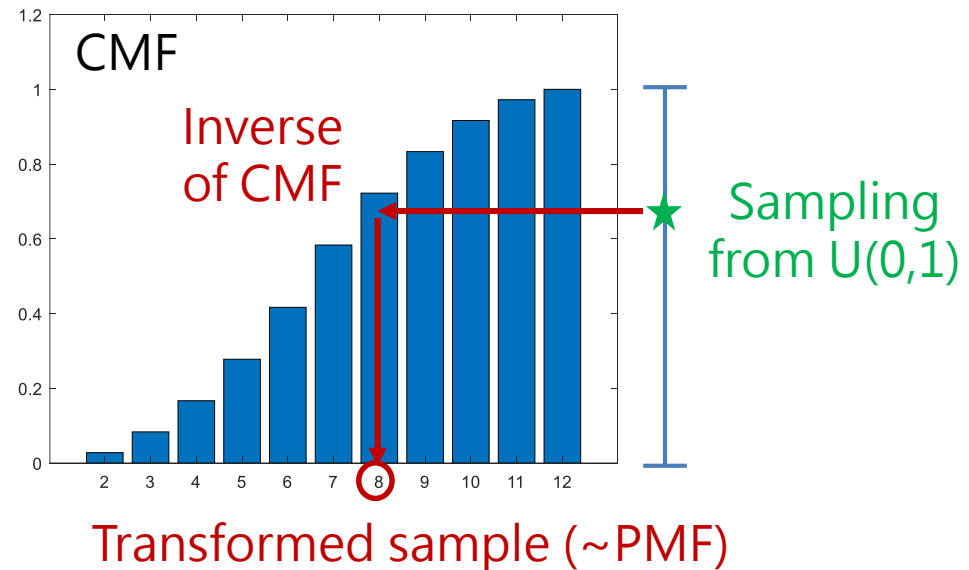
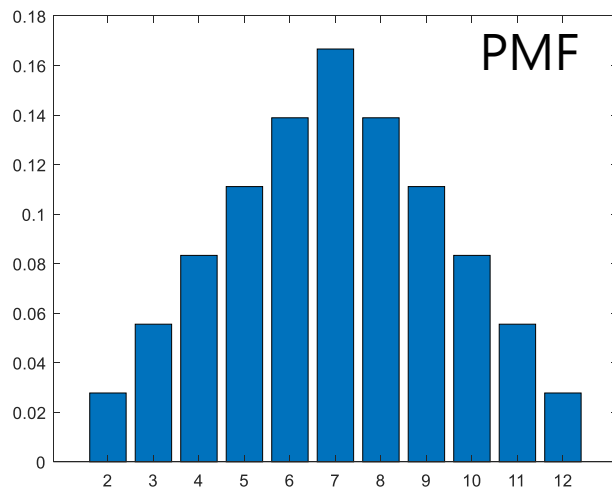
- A learnable distribution transformation model $\mathcal{G}_\theta$ between tractable latent r.v. $\mathbf{z}$ and observable real r.v. $\mathbf{x}$
- A learnable process $\mathbf{x} = \mathcal{G}_\theta(\mathbf{z}, \mathbf{v})$ with latent variable $\mathbf{z}$ (and sometimes another $\mathbf{v}$) to generate distribution $p_\theta(\mathbf{x})$ which is similar to real distribution $q(\mathbf{x})$
 - $\mathbf{z}$: latent variable as stochastic encoding (usually Gaussian)
 - $\mathbf{v}$: latent variables required for stochastic $\mathcal{G}_\theta$



Credit: Sohl-Dickstein et al., ICML'15; DPM.

Examples of distribution transformation

- Generate samples of a fair dice using three fair coins (how?)
- Inverse transform sampling



- Gaussian-camera noise conversion in Noise Flow (via normalizing flow)



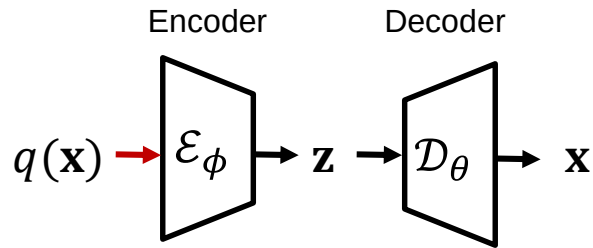
Distinctions of generative models

- On formulation: flexibility vs tractability
 - Flexibility: capability of generating samples $\mathbf{x}$
 - Tractability: capability of evaluating likelihood $\log p_{\theta}(\mathbf{x})$
- On optimization: likelihood-based vs likelihood-free
 - Likelihood-based: usually good tractability with worse flexibility
 - Likelihood-free: usually good flexibility with worse tractability
- On sampling: prescribed vs implicit
 - Prescribed: stochastic $\mathcal{G}_{\theta}$ with $\mathbf{v}$
 - Implicit: deterministic $\mathcal{G}_{\theta}$ without $\mathbf{v}$



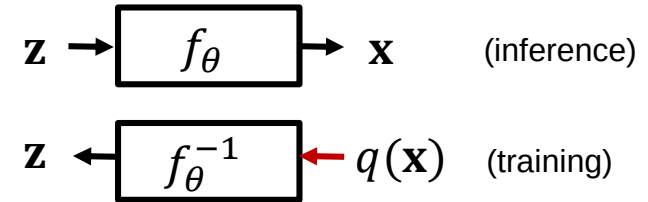
Colored lines
indicate sampling

Popular approaches (1/2)



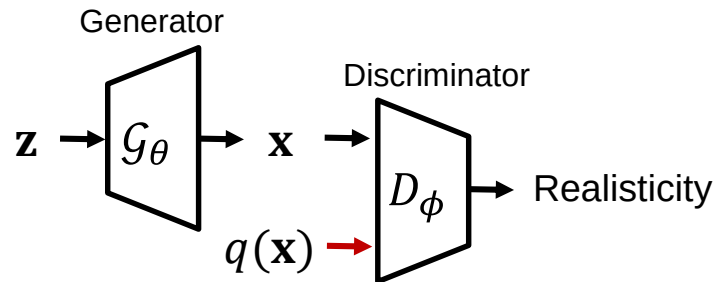
Variational auto-encoder (VAE):

- Implicit
- Likelihood-based
- Likelihood estimated by ELBO via $\mathcal{E}_\phi$



Normalizing flow:

- Implicit
- Likelihood-based (exact with f_θ^{-1})
- Evolves to current SOTA flow matching (FM) in fidelity



Generative Adversarial Network (GAN):

- Implicit
- Likelihood-free with discriminator (intractable likelihood)
- Previous SOTA in fidelity

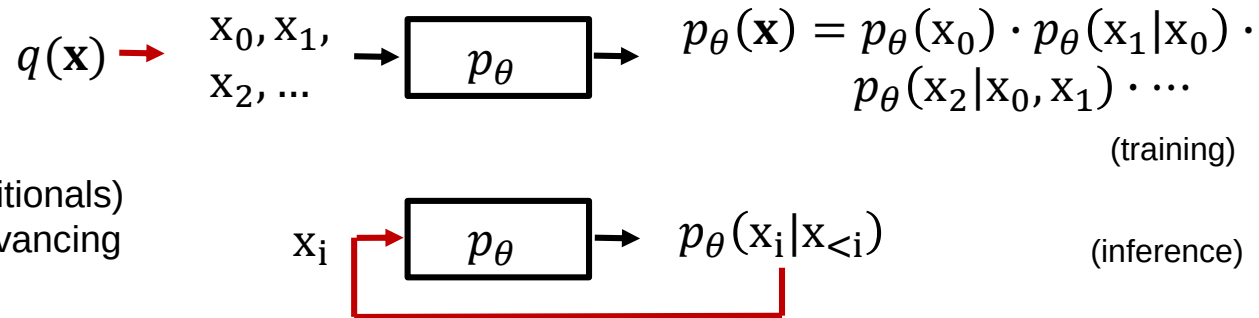


Popular approaches (2/2)

Colored lines
indicate sampling

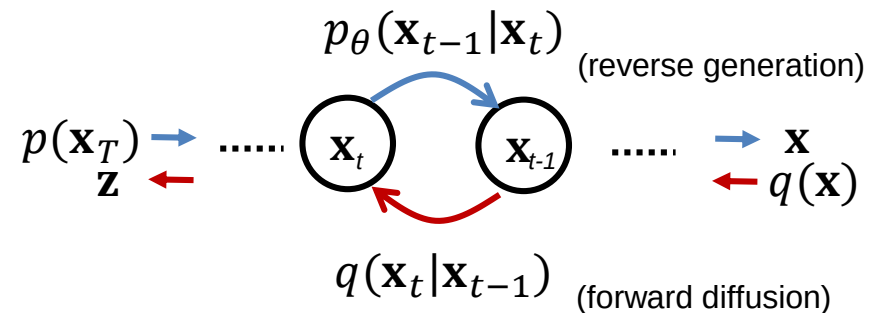
Autoregressive:

- Prescribed
- Likelihood-based (exact by conditionals)
- Previous SOTA on likelihood; advancing towards SOTA fidelity



Diffusion Model (DM):

- Prescribed (DDPM/SDE) or implicit (DDIM/ODE/FM)
- Likelihood-based (ELBO via fixed $q(\mathbf{x}_t|\mathbf{x}_{t-1})$)
- Current SOTA in both fidelity and likelihood



Quality assessment

- Negative log likelihood (NLL) $\downarrow$ (low is good)
 - Model fitness of learned DM w.r.t. real distribution formed by real samples
 - Cross entropy of $p_{\theta}(\mathbf{x})$ over $q(\mathbf{x})$; no generated samples explicitly required
- Inception score (IS) $\uparrow$ (high is good)
 - No-reference metric on generated samples only (no real samples required)
 - Measures generated data distribution w/ classifier
- Fréchet inception distance (FID) $\downarrow$ (low is good)
 - Reference metric in terms of distance between feature distributions of real and generated samples
 - Combine two aspects of image quality, fidelity and diversity, into one value

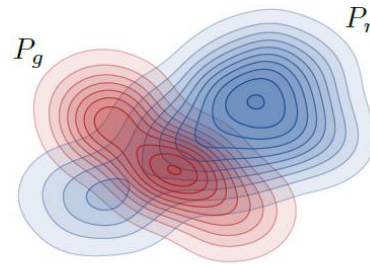
- Fidelity = precision
- Diversity = recall



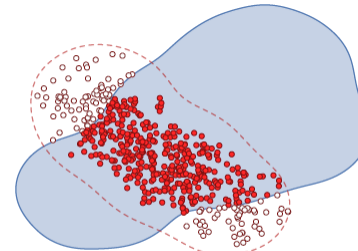
“Corgi”

FID $\nearrow$

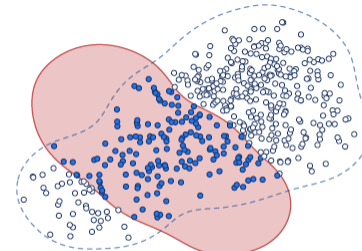
- Can trade diversity for fidelity
 - E.g. give more human instructions/prompts



(a) Example distributions



(b) Precision



(c) Recall

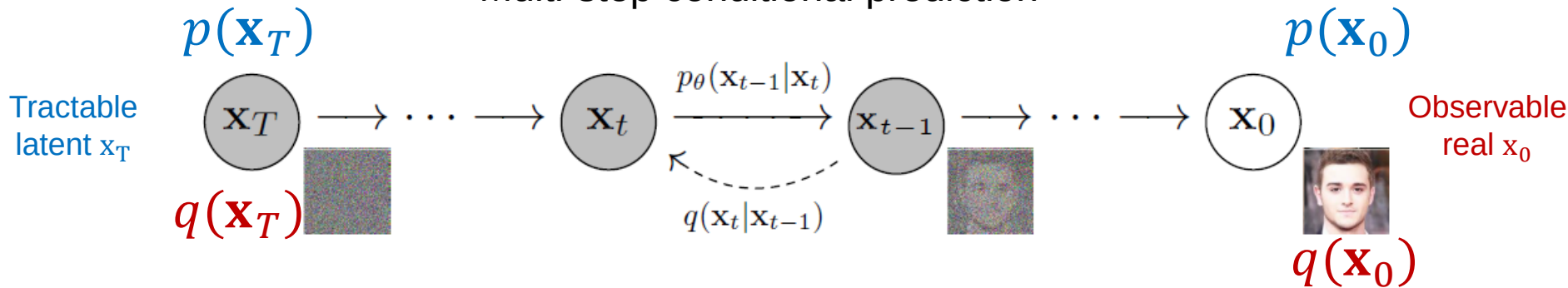
Credit: Kynkäänniemi, NeurIPS'19.



Mainstream framework of diffusion model

Reverse process (learnable; generating $\mathbf{x}_T, \mathbf{x}_{T-1}, \dots, \mathbf{x}_0$):

Multi-step conditional prediction



Forward process (fixed):

Multi-step Gaussian noise addition

$q(\cdot)$ for forward process;
 $p(\cdot)$ for reverse process.

Interchangeable symbols:
 $q(\mathbf{x}_t), q_t(\mathbf{x}_t), q(\mathbf{x}, t)$

- Learnable prediction networks for reverse process
 - Usually implemented by U-Net or Transformer
 - Generate $\mathbf{x}_{t-1}$ from $\mathbf{x}_t$ to satisfy $p(\mathbf{x}_{t-1} | \mathbf{x}_t)$
 - Discrete-time DMs: noise prediction ϵ_θ , data prediction $\mathbf{x}_\theta$, v-prediction $\mathbf{v}_\theta$ for DDPM/DDIM
 - Continuous-time DMs: score prediction of $\nabla_{\mathbf{x}} \log q(\mathbf{x}, t)$ for SDE/ODE, velocity prediction of $\frac{d\mathbf{x}_t}{dt}$ for FM
- Sampling procedure from $t=T$ to 0 for reverse process
 - Sample $\mathbf{x}_T$ from tractable $p(\mathbf{x}_T, T)$
 - Discrete-time DMs: Markov chain updates with prediction results
 - Continuous-time DMs: Markov process updates with prediction results



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Unconditional image synthesis (1/3)

- DDPM sample examples

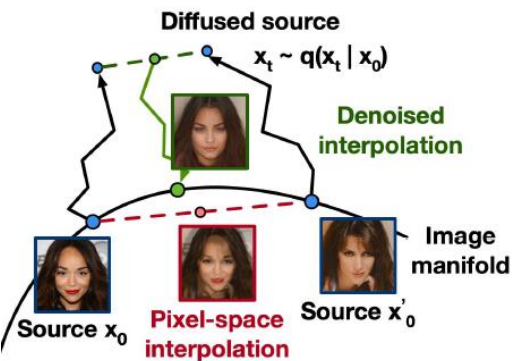


CelebA-HQ 256x256



LSUN Church 256x256

- Interpolation over latent encoding at intermediate time t

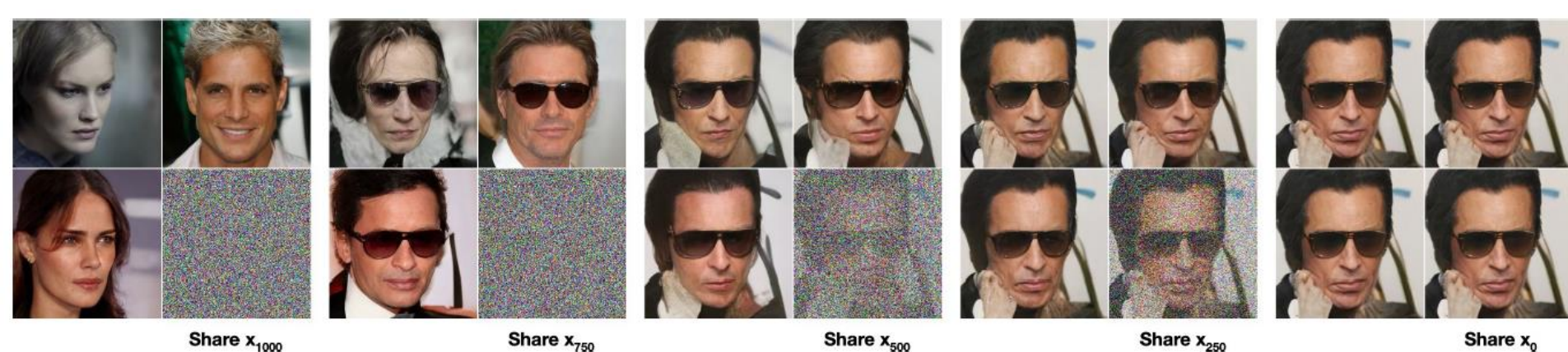


Unconditional image synthesis (2/3)

- Progressive generation
 - Estimates $\hat{\mathbf{x}}_0$ over time
 - Reverse generation from T to t with $p_\theta(\mathbf{x}_t|\mathbf{x}_T)$ and then estimate $\hat{\mathbf{x}}_0(\mathbf{x}_t, t)$ directly



- Sampling from the same latent $\mathbf{x}_t$ (more conceptual for larger t)



Unconditional image synthesis (3/3)

- DMs beat GANs [Dhariwal et al., NeurIPS'21; ADM]

BigGAN-deep



ADM



Conditional/guided image synthesis (1/6)

- Conditioned on $\mathbf{c}$, aim to generate for forward $q(\mathbf{x}|\mathbf{c})$
 - Colorization: $q(\mathbf{x}|\mathbf{x}_0^Y)$ via $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t = \{\mathbf{x}_t^Y, \mathbf{x}_t^{UV}\})$ and $q(\mathbf{x}_t^Y|\mathbf{x}_0^Y)$
 - $\mathbf{x}_t^Y$: luma as condition $\mathbf{c}$, $\mathbf{x}_t^{UV}$: chroma as unknown
 - No retraining



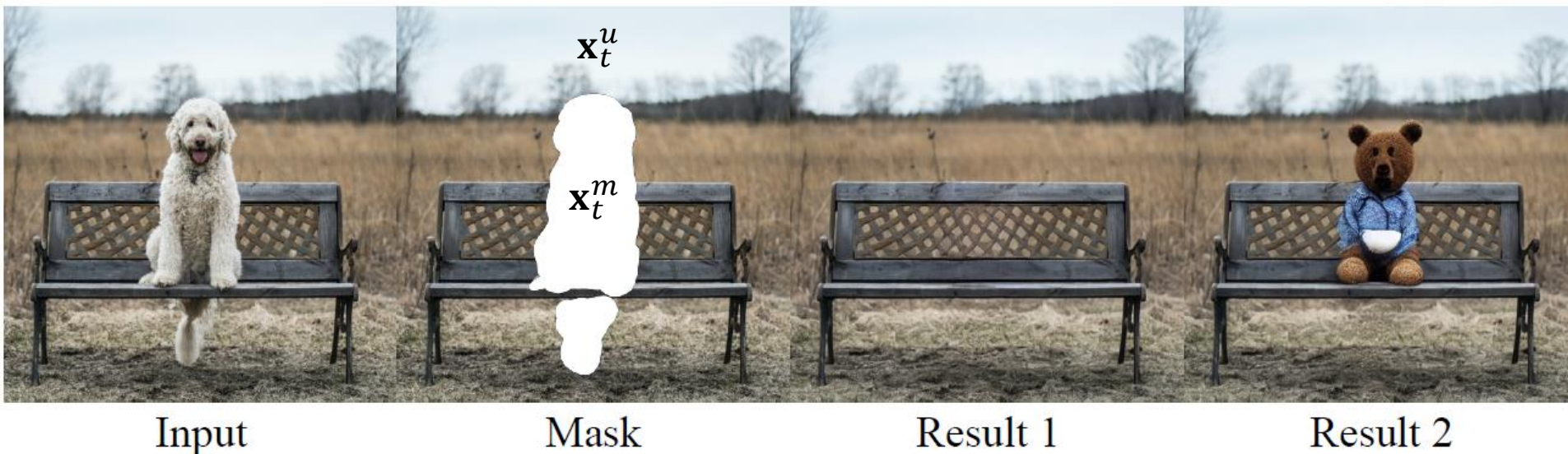
Original

Grey input

Credit: Song et al., ICLR'21; SDE.

Conditional/guided image synthesis (2/6)

- Inpainting: $q(\mathbf{x}|\mathbf{x}_0^u)$ via $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t = \{\mathbf{x}_t^u, \mathbf{x}_t^m\})$ and $q(\mathbf{x}_t^u|\mathbf{x}_0^u)$
 - $\mathbf{x}_t^u$: unmasked as condition $\mathbf{c}$, $\mathbf{x}_t^m$: masked as unknown
 - No retraining



Credit: Meng et al., CVPR'23; GDM.

Conditional/guided image synthesis (3/6)

- Class-guided synthesis: $\nabla_{\mathbf{x}} \log q(\mathbf{x}|\mathbf{c}, t) = \nabla_{\mathbf{x}} \log q(\mathbf{x}, t) + \nabla_{\mathbf{x}} \log q(\mathbf{c}|\mathbf{x}, t)$
 - $\mathbf{c}$: class (e.g. Corgi)
 - Train another classifier for $q(\mathbf{c}|\mathbf{x}, t)$ or classifier-free guidance ($\sim \nabla_{\mathbf{x}} \log q(\mathbf{c}|\mathbf{x}, t)$)
 - Trade diversity for fidelity



Non-guided

Classifier-free-guided

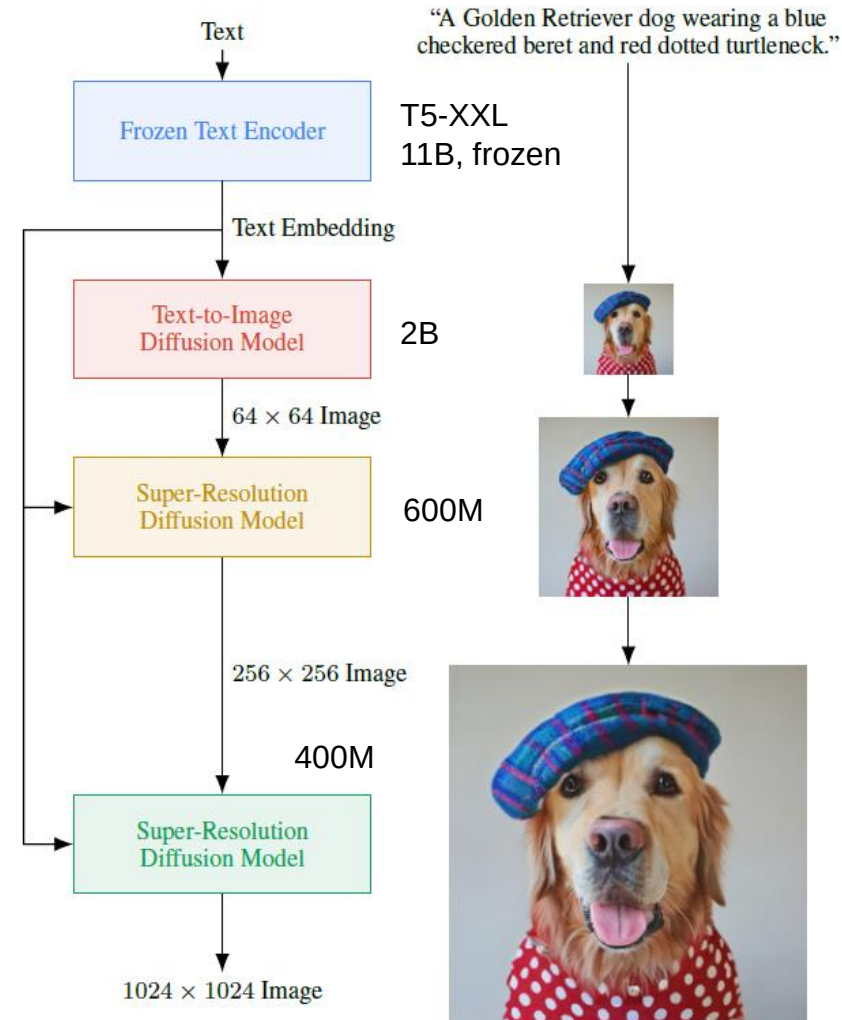
Credit: Ho et al., NeurIPS'21.

Conditional/guided image synthesis (4/6)

- Text-to-image: $\nabla_{\mathbf{x}} \log q(\mathbf{x}|\mathbf{c}, t)$
 - $\mathbf{c}$: text embedding from language model (LM)
 - Train DM for $q(\mathbf{x}|\mathbf{c}, t)$ with text embedding $\mathbf{c}$
 - Examples from Imagen:



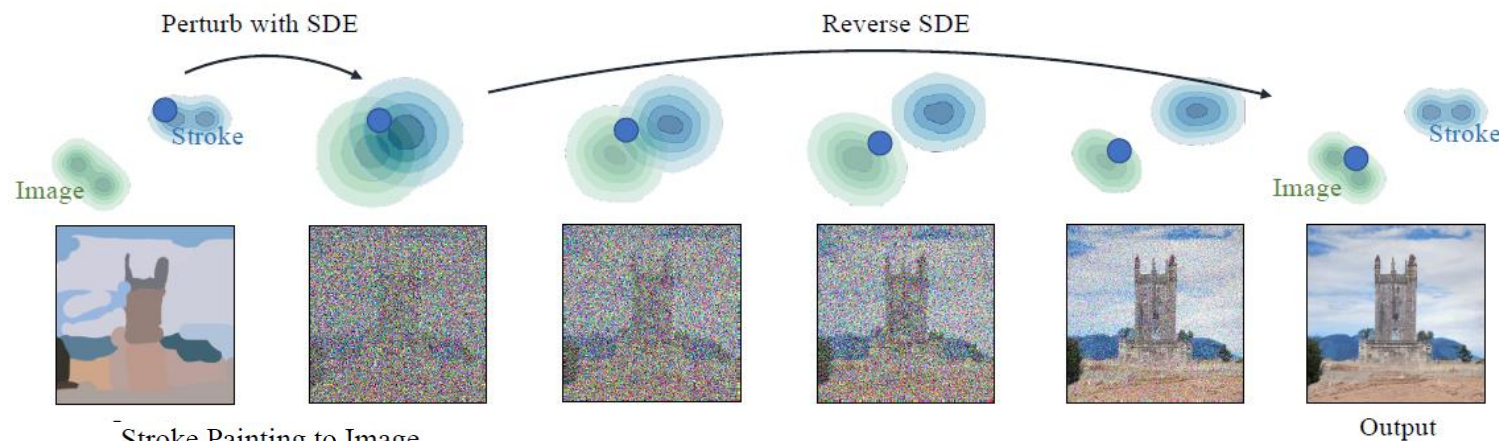
A cute corgi lives in a house made out of sushi.



Credit: Saharia et al., NeurIPS'22; Imagen.

Conditional/guided image synthesis (5/6)

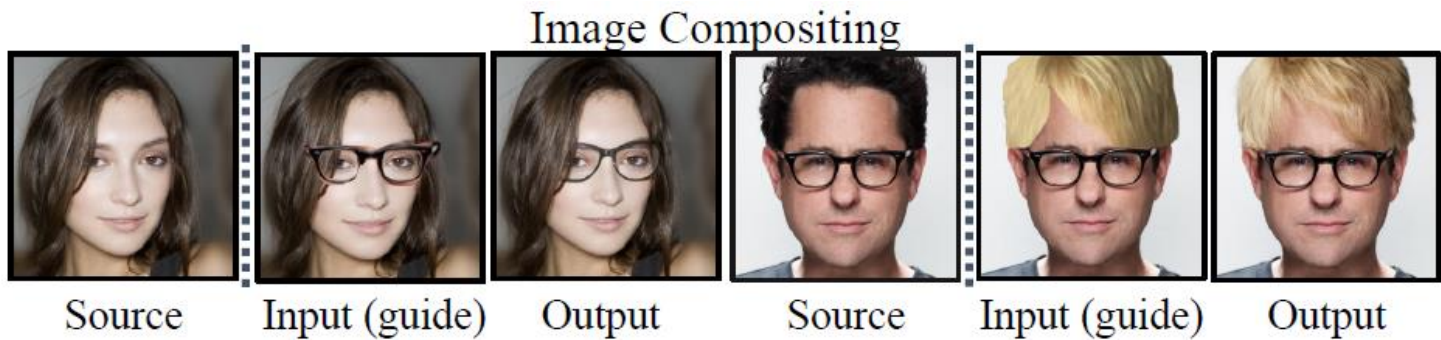
- Image-to-image translation: $\nabla_{\mathbf{x}} \log q(\mathbf{x}|\mathbf{c}, t)$
 - $\mathbf{x}_0^G$: clean guided image as condition $\mathbf{c}$
 - Either direct training for $\nabla_{\mathbf{x}} \log q(\mathbf{x}|\mathbf{c}, t)$
 - Or without training, e.g. SDEdit with $\nabla_{\mathbf{x}} \log q(\mathbf{x}, t)$ from t_0 with $q(\mathbf{x}_{t_0}^G | \mathbf{x}_0^G)$



Credit: Meng et al., ICLR'22; SDEdit.

Conditional/guided image synthesis (6/6)

- More Image-to-image translation examples



Credit: Meng et al., ICLR'22; SDEdit.



Credit: Meng et al., CVPR'23; GDM.

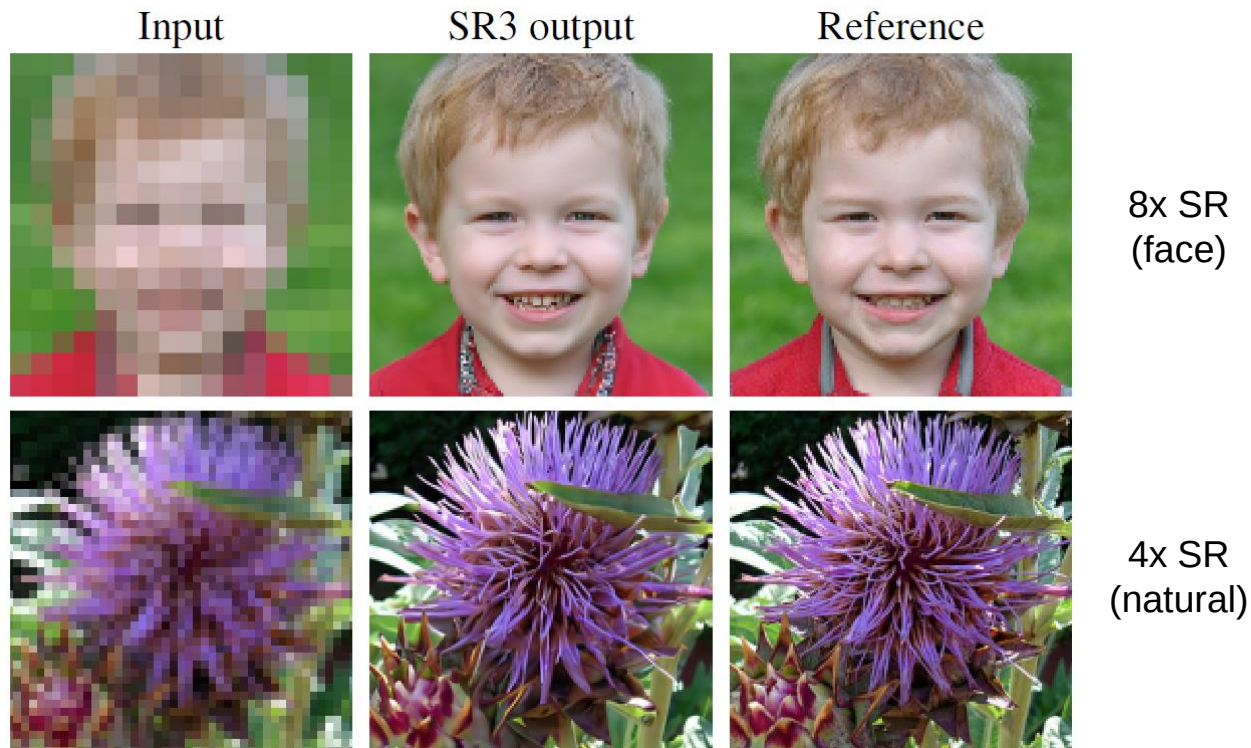


Video generation (e.g. SVD)

Stable Video Diffusion

Super resolution

- Given low-resolution $\mathbf{x}^{\text{LR}}$, train DM for $q(\mathbf{x}|\mathbf{x}^{\text{LR}})$
 - SR3 [TPAMI'23]: embed bicubic-upsampled $\mathbf{x}^{\text{LR}}$ into U-Net
 - SRDiff [Nerocomputing'22]: embed SR of $\mathbf{x}^{\text{LR}}$ into U-Net



Close-up image

Bicubic



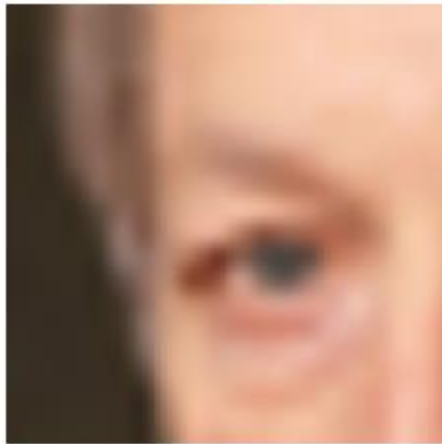
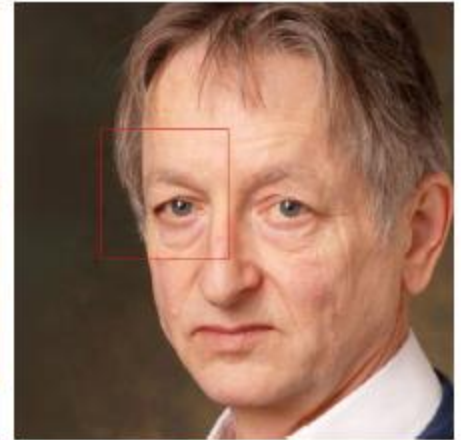
Regression



SR3 (ours)

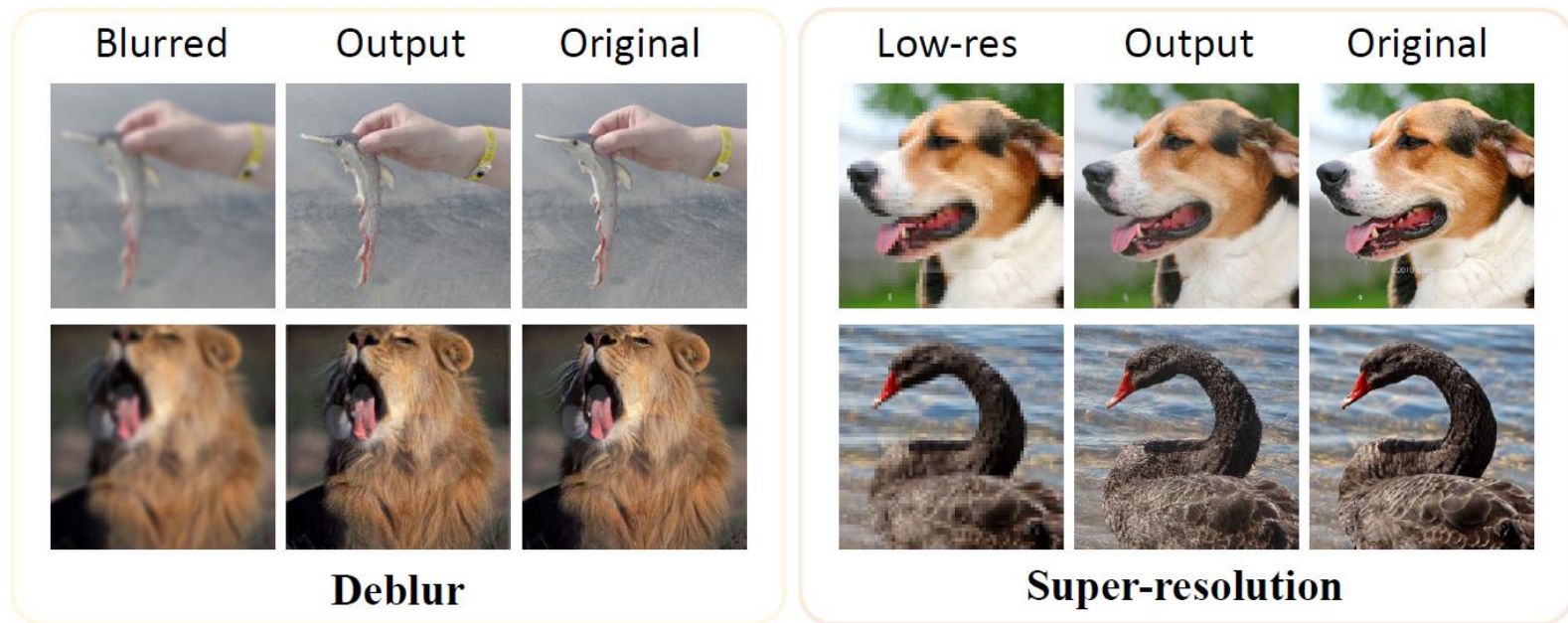


Reference



DM for unified image restoration and enhancement

- Given distorted $\mathbf{y}$, construct optimization framework with image prior $q(\mathbf{x})$ as regularization term that is implemented by score $\nabla_{\mathbf{x}} \log q(\mathbf{x}, t)$
 - GDP [CVPR'23]: diffusion with heuristic guidance from reconstruction loss (data term with $\mathbf{y}$) and quality loss (regularization term on $\mathbf{x}$)



Credit: Fei et al., CVPR'23; GDP.



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 - **Quality metrics: NLL, IS, FID**
 - **DM as image prior**



Negative Log Likelihood (NLL) (1/2)

- $NLL \triangleq \mathbf{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0)} [-\log p_\theta(\mathbf{x}_0)]$
 - Cross entropy of the generative distribution $p_\theta(\mathbf{x}_0)$ over the real $q(\mathbf{x}_0)$
 - Defined on the real dataset $\{\mathbf{x}_0\}$ (NOT related to generated pixels)
 - Evaluate the information amount if we transmit or compress a source of $q(\mathbf{x}_0)$ with the trained $p_\theta(\mathbf{x}_0)$
 - Lower bound is the source entropy $\mathbf{E}_{q(\mathbf{x}_0)} [-\log q(\mathbf{x}_0)]$
 - Faster samplers usually degrade NLL significantly (while maintaining FID)
 - Minimal random code is a method that is able to send samples drawn from $q(\mathbf{x}_0)$ with NLL-nats on average if the receiver has $p_\theta(\mathbf{x}_0)$ w/o $q(\mathbf{x}_0)$
- Two difficulties for evaluation:
 - 1) Source pixels have discrete values (without explicit continuous $q(\mathbf{x}_0)$)
 - 2) In some cases, it is hard to exactly evaluate $\log p_\theta(\mathbf{x}_0)$
 - DM and VAE can use ELBO to estimate an upper bound, but GAN cannot
 - (while normalizing flow and AR have exact forms)



Negative Log Likelihood (NLL) (2/2)

- Solutions for evaluation:

- 1) Uniform dequantization for the discrete-value issue

- Define noisy data $\mathbf{y} = \mathbf{x} + \mathbf{u}$, $\mathbf{u} \in U(0,1)^D$ for clean data $\mathbf{x} \in [0,1, \dots, 255]^D$
 - Convert the pmf $Q(\mathbf{x})$ of discrete-valued $\mathbf{x}$ into a pdf of continuous pdf $q(\mathbf{y})$
 - Re-define $NLL \triangleq \mathbf{E}_{\mathbf{y} \sim q(\mathbf{y})} [-\log p_\theta(\mathbf{y})]$
 - Estimate NLL with i.i.d. samples of $\mathbf{x}$ and $\mathbf{u}$: $\frac{1}{N} \frac{1}{M} \sum_{i=1}^N \sum_{j=1}^M -\log p_\theta(\mathbf{x}^{(i)} + \mathbf{u}^{(j)})$

- 2) Estimation with variational upper bound (RHS); example for DM:

- $$-\log p_\theta(\mathbf{x}_0) \leq D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))$$
$$+ \sum_{t=2}^T \mathbf{E}_{q(\mathbf{x}_t)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$
$$+ \mathbf{E}_{q(\mathbf{x}_1)} [-\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$
 - Estimate with many samples of latent variables $\mathbf{x}_{1:T}$ which are sampled from forward diffusion process $q(\mathbf{x}_t|\mathbf{x}_{t-1})$
 - Note: This method over-estimates NLL, and the actual NLL is smaller.



Inception score (IS)

- A no-reference metric which directly measures the generated distribution $p_{\theta}(\mathbf{x}_0)$ with
 - Classification network output, particularly Inception v3: $p(c|\mathbf{x})$
 - Distribution of the results on generated data: $p(c; \theta) = \int p(c|\mathbf{x})p_{\theta}(\mathbf{x})d\mathbf{x}$

$$\log \text{IS}[\mathbf{x}] = \mathbf{E}_{\mathbf{x} \sim p_{\theta}(\mathbf{x})} [D_{KL}(p(c|\mathbf{x}) \parallel p(c; \theta))]$$

$$= \underbrace{H(p(c; \theta))}_{\substack{\uparrow \text{ for more diversity} \\ \text{with higher entropy}}} - \mathbf{E}_{\mathbf{x} \sim p_{\theta}(\mathbf{x})} [\underbrace{H(p(c|\mathbf{x}))}_{\substack{\downarrow \text{ for more discriminative with} \\ \text{lower conditional entropy}}}]$$

- Evaluated by

$$\text{IS}[\mathbf{x}] = \exp \left(\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K p(c_k | \mathbf{x}^{(i)}) \log \frac{p(c_k | \mathbf{x}^{(i)})}{p(c_k; \theta)} \right)$$



Fréchet inception distance (FID)

- Fréchet distance ($\equiv$ Wasserstein-2 distance)
 - Distance $d_F(p, q)$ between two pdf's p and q of $\mathbf{x} \in \mathbb{R}^n$

$$d_F(p, q) = \left(\inf_{\gamma \in \Gamma(p, q)} \int_{\mathbb{R}^n \times \mathbb{R}^n} \|\mathbf{x} - \mathbf{y}\|^2 d\gamma(\mathbf{x}, \mathbf{y}) \right)^{1/2}$$

- Squared Fréchet distance between two Gaussians $p \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $q \sim N(\boldsymbol{\mu}', \boldsymbol{\Sigma}')$:

$$d_F^2(p, q) = \|\boldsymbol{\mu} - \boldsymbol{\mu}'\|^2 + \text{tr}(\boldsymbol{\Sigma} + \boldsymbol{\Sigma}' - 2(\boldsymbol{\Sigma}\boldsymbol{\Sigma}')^{1/2})$$

- FID is the squared Fréchet distance
 - Between two distributions of the pool layer (2048x1) of Inception v3 on two compared datasets, e.g. real vs generated
 - Assume that these two distributions are Gaussian, and, therefore, FID can be determined by their means and variances.



DM as image prior

- Given condition $\mathbf{y}_0$, the conditional probability over time becomes
 - The condition can be a given class or image

$$q(\mathbf{x}_t|\mathbf{y}_0) = q(\mathbf{y}_0|\mathbf{x}_t) \cdot q(\mathbf{x}_t)/q(\mathbf{y}_0)$$

- Then we can generate images with the guided score:

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{y}_0) = \underbrace{\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t)}_{\text{pre-trained score as prior}} + \nabla_{\mathbf{x}_t} \underbrace{\log q(\mathbf{y}_0|\mathbf{x}_t)}_{\text{data term for image restoration (hard to directly evaluate)}}$$

pre-trained score
as prior

data term for image restoration
(hard to directly evaluate)